

Business and Technical Summary

Quantum Optimization for Distributed Order Management — WISER × Nestlé Challenge

What is Distributed Order Management (DOM)?

Distributed Order Management is the process of deciding which distribution center (DC) fulfills each customer order across a network. Most orders are handled by a default DC, but when that DC cannot fully satisfy an order — due to limited inventory, dock capacity, or throughput — planners must decide whether to divert it to an alternative DC, fulfill it partially, or accept a penalty for unmet demand. The objective is to make these reassignment decisions so as to maximize overall fulfillment value while controlling shipping cost and penalty exposure.

Why DOM is a Combinatorial Optimization Problem

Each order can potentially be assigned to any of several candidate DCs, or left partially or fully unfulfilled. Every DC draws on shared, finite resources — inventory per SKU, and in some cases dock or throughput limits — that many orders compete for at the same time. The number of possible assignment combinations grows exponentially with the number of orders and DCs, which is the defining feature of a combinatorial optimization problem. Critically, the best assignment for one order depends on what other orders have already consumed from shared inventory, so the problem must be solved jointly across all orders rather than one order at a time.

Why Advanced Optimization Methods Are Relevant

Because the search space grows exponentially, exhaustively enumerating every assignment combination is only feasible for very small instances. Methods purpose-built for large combinatorial search — binary integer programming, weighted exact-cover / column-selection formulations, and quantum or quantum-inspired approaches such as QAOA — are relevant because they can either guarantee (or bound) solution quality efficiently, or find strong approximate solutions when an exact search is computationally out of reach.

Key Trade-offs

Three trade-offs shape how a DOM solver should be designed and evaluated:

- Solution quality vs. runtime — exact solvers can guarantee optimality but scale poorly as the number of orders and DCs grows; heuristics and approximate methods sacrifice some solution quality for speed, which matters when reassignment decisions must be made quickly in live operations.
- Exact solvers vs. heuristics — exact methods (e.g., ILP solvers) give provably optimal or bounded-gap solutions but can become intractable at scale; greedy or sequential heuristics scale well but offer no optimality guarantee and can lock in poor early decisions.
- Noisy quantum hardware vs. simulators — current quantum hardware has limited qubit counts and is subject to noise, which constrains problem size and solution reliability; simulators avoid noise entirely but do not reflect real hardware constraints, so simulator results should be read as a best-case proxy rather than a production-ready benchmark.

Trade-off	Favors A	Favors B
Solution quality vs. runtime	Exact solvers: optimal, but scale poorly	Heuristics: fast, but no optimality guarantee
Exact solvers vs. heuristics	ILP/exact-cover: provable bounds	Greedy/sequential: scalable, may get stuck
Noisy hardware vs. simulators	Real QPU: reflects actual noise/limits	Simulator: clean, best-case proxy only

Relevance to Nestlé-style Operations

For a global food and beverage operation with many distribution centers, small improvements in reassignment decisions compound across thousands of daily orders — reducing unmet demand and shipping cost while respecting real inventory and capacity limits. A transparent classical baseline paired with a rigorously benchmarked advanced method gives planners both a trustworthy decision and a clear account of why it outperforms simpler rules.